

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2024-2025****EE6402 – REAL-TIME DSP DESIGN AND APPLICATIONS**

April / May 2025

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. (a) Consider an infinite precision signal $w(n)$ being truncated to the binary fixed point representation $[w(n)]$ in Q3.4 format.
 - (i) What is the probability density function of the truncation error $e(n) = [w(n)] - w(n)$ in the above representation?
 - (ii) What is the mean of $e(n)$?
 - (iii) How many bits are needed to represent a number in Q3.4 fixed point format?
 - (iv) What is the dynamic range of Q3.4 fixed point format?
 - (v) What is the precision of Q3.4 fixed point format?
 - (vi) Express the decimal number $(-5.625)_{10}$ in binary Q3.4 fixed point format.

(14 Marks)

Note: Question No. 1 continues on page 2.

- (b) An input $x(n)$ passes through a 100-tap FIR filter with frequency response $H(e^{j\omega}) = \begin{cases} 1, & \frac{\pi}{2} \leq |\omega| \leq \pi \\ 0, & \text{otherwise} \end{cases}$ to give an output $y(n)$ (see Figure 1). The input $x(n)$ is rounded using a quantization step size of Q , which may be modelled as a noise injection $e_x(n)$ as shown in Figure 1. After each multiplication, each partial product is truncated to a word-length which is 5 bits longer than the input signal word-length. The truncations are modelled as noise injections $e_i(n)$, $i = 0$ to 99. The output $y(n)$ is also rounded using the same quantization step size of Q , which is modelled as another noise injection $e_y(n)$.

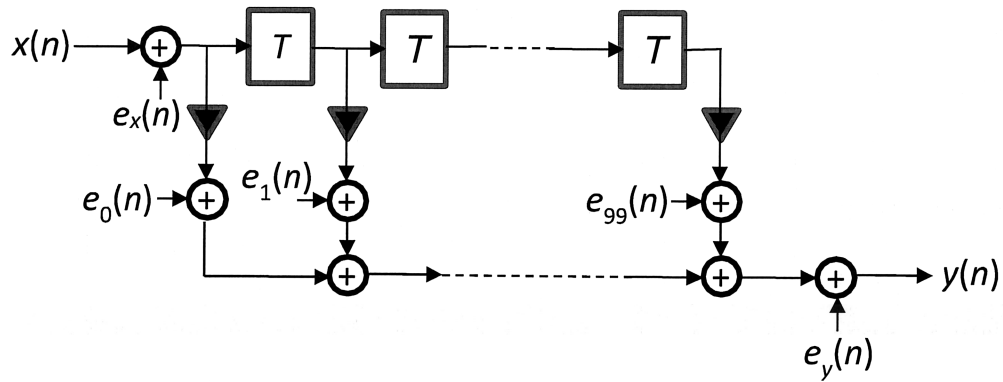


Figure 1

- Find the total noise mean η_{total} (as a function of Q) at the output arising from $e_x(n)$, $e_i(n)$, and $e_y(n)$.
- Find the total noise variance σ_{total}^2 (as a function of Q) at the output arising from $e_x(n)$, $e_i(n)$, and $e_y(n)$.
- If the input $x(n)$ is rounded to Qm.4 format (with 4 fractional bits after the binary point), determine the value of Q . Using your answer, evaluate η_{total} and σ_{total}^2 you found in parts (i) and (ii).
- If the input $x(n)$ is truncated instead of rounding (with the same quantization step size as in part (iii)), what would be η_{total} and σ_{total}^2 ?

(11 Marks)

2. (a) $x_1(t)$, a continuous-time cosine signal of frequency f_1 Hz, $0 \leq f_1 \leq 75$ Hz, is sampled at the rate of 150Hz. $x_2(t)$, another continuous-time cosine signal of frequency 240 Hz, is also sampled at the rate of 150Hz. If both signals yield identical samples, find f_1 . Further, $x_3(t)$, a third continuous-time cosine signal of frequency $m \times 240$ Hz (where m is a positive integer), also yields identical samples. Find the minimum value of m . (6 Marks)
- (b) With the help of a diagram, explain how a parallel (flash) ADC (analog to digital converter) works. In which application areas is a flash ADC most commonly used, and what are the reasons for its preference? (8 Marks)
- (c) The reservation table for a pipelined circuit consisting of 2 stages is shown in Figure 2, where 'A' denotes the stage/resource is being used.

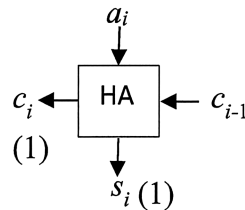
		time			
		0	1	2	3
stages	1	A			A
	2		A	A	

Figure 2

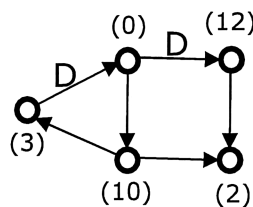
- (i) What is the maximum achievable throughput for this reservation table?
- (ii) Determine the forbidden set of initiations (that causes a collision). Hence, suggest a scheduling strategy (a permissible sequence that does not cause any collision).
- (iii) Find the latency and the average throughput for your scheduling strategy in part (ii).

(11 Marks)

3. (a) A rounding adder is to be designed for rounding a 4 bit input $a_3a_2a_1a_0$ using the carry in bit $c_{-1} = a_{-1}$. The output is $s_3s_2s_1s_0$ and the carry out bit c_3 .
- (i) Using half adders (HA) shown in Figure 3, design the above rounding adder with ripple carry (carry propagates). Draw the block diagram of your design. Both outputs of the half adder need 1 gate delay as shown in the figure. Find the required gate delays for each of the output bits c_3, s_3, s_2, s_1, s_0 .

**Figure 3**

- (ii) Describe how carry look-ahead may be used to reduce the gate delay for c_3 (you may ignore the gate delays for s_3, s_2, s_1, s_0). Draw the block diagram of the 4-bit carry look-ahead rounding adder, and find the required gate delay for c_3 .
- (10 Marks)
- (b) The data flow graph of a system is shown in Figure 4. Each node is labelled by its computation time. If any edge has a delay, it is labelled by D.

**Figure 4**

- (i) Find a loop in the data flow graph. Find the loop bound for this loop.
- (ii) Find the critical path and the computation time it requires.
- (iii) Find a cutset such that cutset retiming reduces the critical path. Draw the retimed data flow graph and show the new critical path.
- (10 Marks)
- (c) The SATD bit (saturation mode bit for D unit) and the SMUL bit (saturation on multiplication bit) in TMS320C55x may be used to saturate the result of an arithmetic operation during overflow. Discuss why saturation is preferred compared to overflow in typical signal processing applications.
- (5 Marks)

4. (a) Figure 5 shows a system. Each node is labelled by its computation time. Each edge is labelled by the number of delays.

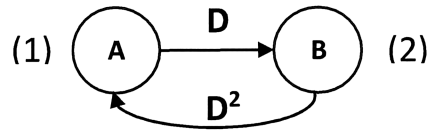


Figure 5

- (i) Find the critical path, and hence find the lower bound on T_S , the sampling period of this system.
- (ii) Use unfolding to replicate the system of Figure 5 for an unfolding factor of $J = 2$. Find the critical path, and hence find the lower bound on T_S after unfolding.
- (11 Marks)
- (b) (i) In TMS320C55x, using the rules for moving a value to a register, find the value of the auxiliary register AR0, in hexadecimal, after each of the following instructions.
- ```

MOV #10, AR0
MOV #Ah, AR0
MOV #-10, AR0
MOV #12345, AR0
MOV #1234h, AR0

```
- (ii) If the decimal value  $(-250)_{10}$  or its hexadecimal equivalent is to be moved to AR0, write a MOV instruction (in mnemonic form) to achieve this.
- (14 Marks)

END OF PAPER





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EE6402 REAL-TIME DSP DESIGN & APPLICATIONS

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.